# ACoP 2026 Hackathon: detailed guide

Beyond CTG Repeats: Advancing Patient Stratification in Myotonic Dystrophy Type 1 (DM1) | Team 1 | companion to the three-page brief | 11 October 2026

**How to read this.** Statements from the kickoff slides or session are given plainly. Boxes in green are the author's own reading or suggestion, and should be checked by the team. The dataset is simulated; nothing here describes real patients.


1. What we are being asked
2. The disease, step by step
3. Why patients differ
4. The three treatments
5. The dataset, column by column
6. The approach, step by step
7. The methods in plain words
8. Reading the results
9. Pitfalls and likely judge questions
10. Logistics and scoring
11. Glossary


## 1. What we are being asked

**The hackathon question:** can machine learning identify additional predictors and improve patient stratification beyond traditional covariate modelling?

We receive a simulated clinical trial of 500 DM1 patients. Each patient received one of three treatments, and for each treatment there is a measure of how well the patient did. The job has two parts:

1. **Identify clinically meaningful covariates per treatment arm.** A covariate is a baseline patient characteristic such as age or muscle strength. "Meaningful" means it really predicts response and makes biological sense for that drug.
2. **Compare covariate models with machine-learning models using proper metrics.** Fit both kinds, score them fairly, and say whether the ML found anything the traditional methods missed.

The organizers stress one line three times in the deck: **"Meaningful covariate identification matters more than fancy models."** The exercise is a test of judgement, not of who builds the most complicated model.

"Stratification" means sorting patients into groups that behave alike, here groups likely or unlikely to respond to a given drug. It matters for trial design (whom to enrol) and for treatment choice (who benefits).

## 2. The disease, step by step

Myotonic dystrophy type 1 is an inherited disease. The name describes two muscle features: *myotonia*, muscle that is slow to relax, and *dystrophy*, muscle that wastes over time. It is rare by rate, about 16.7 per 100,000 people (1 in 6,000), but that still means an estimated 56,500 patients in the United States.

**Step 1. A stutter in one gene.** A gene called *DMPK* on chromosome 19 contains the three DNA letters C-T-G repeated in a row. Everyone has some repeats: 5 to 37 is normal. In DM1 the count is expanded, from 50 to several thousand.

**Step 2. The stutter is copied into RNA.** To use a gene, the cell first copies it into RNA. The long repeat (written CUG in RNA) makes the copy fold into a sticky clump that accumulates inside the cell nucleus. This is called toxic RNA.

**Step 3. The clump traps an editor protein.** Proteins called MBNL normally edit other RNA messages so the correct version of each protein is produced. The editing is called splicing. The toxic RNA soaks up MBNL, and a second regulator, CELF1, becomes overactive.

**Step 4. Many proteins come out wrong.** With the editor missing, many RNA messages are spliced incorrectly, and cells produce faulty versions of the corresponding proteins.

**Step 5. Many organs are affected.** Which organ suffers depends on which protein is faulty there.

| Mis-spliced message | Consequence | What the patient experiences |
|---|---|---|
| Chloride channel in muscle | Myotonia | Grip that will not release; stiffness |
| Calcium channel and structural proteins in muscle | Myopathy (weakness, wasting) | Weak grip, foot drop, gradual loss of muscle |
| Sodium channel in the heart | Arrhythmia, conduction delay | Slow or irregular heartbeat; a serious risk |
| Insulin receptor | Insulin resistance | Tendency to diabetes |
| Brain proteins | Cognitive impairment | Poor concentration, daytime sleepiness |
| Lens proteins | Cataracts | Early clouding of vision |
| Breathing and gut muscle | Respiratory and gastrointestinal problems | Weak breathing, disturbed sleep, swallowing trouble, constipation |

**Myotonia and weakness are different.** Myotonia is a muscle that cannot switch off. Weakness is a muscle that cannot produce force. A patient may have much of one and little of the other, and a drug that helps one may not help the other.

## 3. Why patients differ

- **Repeat count and severity.** Larger expansions broadly go with earlier onset and worse disease. The slide's bands: 38 to 49 premutation (no disease); 50 to 100 mild or late onset; 50 to 1,000 classic adult onset; 800 or more childhood onset; 1,000 or more congenital onset.
- **But the count does not predict an individual well.** The slide title says individual progression "remains unpredictable due to tissue-specific mosaicism". The repeat keeps growing during life, and at different speeds in different tissues, so the number measured in blood is not the number in muscle or heart. This is *somatic mosaicism*.
- **Inheritance.** DM1 is autosomal dominant: one expanded copy from either parent is enough, and each child of an affected parent has a 50% chance of inheriting it. The repeat often grows when passed on, so children can be affected earlier and more severely than the parent. This is called *anticipation* and is particularly associated with transmission from the mother in the congenital form. Siblings who both inherit it can have different repeat lengths and different severity.
- **Different dominant problems.** Two patients with the same genetic fault can have different leading manifestations, one mainly cardiac, another mainly muscular.

The organizers' conclusion: "Disease severity, CTG repeat instability, age, and tissue-specific RNA-processing effects may contribute to variability in treatment response." That sentence is effectively the list of things to look for.

## 4. The three treatments

There are 19 active DM1 development programmes: 1 sodium channel blocker, 11 targeting *DMPK*, 1 GSK3-beta inhibitor and 6 with other targets. The dataset takes one from each of the first three classes. They act at different points in the disease chain, which is the key to interpreting predictors.

### Mexiletine: symptomatic

- **What it does.** Blocks sodium channels and so reduces the abnormal repetitive firing (myotonic discharges) of muscle. The session's image: "turning down the volume of an over-excited muscle". It treats the symptom and does nothing about the genetic cause.
- **Outcome in the dataset.** `myotonia_change_12wk`: change in myotonia severity at week 12. **Negative means improvement.**
- **In real trials.** Hand-grip relaxation time and patient-reported stiffness. A six-month placebo-controlled DM1 trial found no significant change in six-minute walk distance, so relief of myotonia does not always become functional gain.
- **Hint given.** "The drug acts on muscle excitability, so clinical presentation may matter. Some clinical features may be informative, while others may not."

### Oligonucleotide: disease-modifying

- **What it does.** An antisense oligonucleotide is a short synthetic strand that binds the mutant *DMPK* RNA so that the cell destroys it. Less toxic RNA frees the trapped MBNL, and normal splicing can resume. It acts at the root of the disease.
- **Outcome in the dataset.** `qmt_change_26wk`: change in the Quantitative Muscle Testing composite at week 26. QMT is a measured muscle-strength score. **Positive means improvement.**
- **In real trials.** QMT is usually paired with video hand opening time (vHOT, a myotonia measure), timed walk tests, patient-reported indices and splicing biomarkers. vHOT is a common primary endpoint but varies widely between patients (baseline 5.9 ± 7.0 seconds in 591 patients), which is why this dataset uses QMT.
- **Hint given.** "The drug acts on RNA, so molecular and clinical burden may both matter. Not every disease marker is equally informative."

### Tideglusib: dual-pathway

- **What it does.** Irreversibly inhibits GSK3-beta, an enzyme that is overactive in DM1. Blocking it affects two sets of pathways: Wnt/beta-catenin signalling, and the regulation of protein synthesis and RNA-binding protein activity. Because the enzyme is active in several tissues, the drug suits the multisystem nature of the disease.
- **Outcome in the dataset.** `tideglusib_composite`: a combined cognitive and motor improvement score from 0 to 20 at week 26. **Higher is better.**
- **In real trials.** The congenital DM1 trial REACH-CDM used a clinician rating scale (CDM1-RS) as primary endpoint. It missed that endpoint but showed benefit on several functional assessments.
- **Hint given.** "The target pathway is active in several tissues, so more than one domain may matter. Features from different domains may each contribute."

**Author's reading of the hints (to be tested, not assumed).** Mexiletine: baseline `myotonia_severity`, possibly `phenotype`. Oligonucleotide: `ctg_repeats`, `somatic_mosaicism_index`, `muscle_strength`, possibly `disease_duration`. Tideglusib: `cognitive_function` together with a motor measure such as `muscle_strength`. Writing these down before looking at results lets the team say which expectations the data confirmed and which it did not.

**Why comparing treatments is hard** (from the slides): each trial uses its own endpoint, time point and population, and each drug is compared with its own placebo, not with other drugs. The same average shift can hide a wide spread between patients. In this dataset the three outcomes are on different scales and in different directions, so **results cannot be compared across arms**, only within an arm.

## 5. The dataset, column by column

500 simulated patients, one row each, 26 columns: 18 baseline covariates, three continuous outcomes, three responder labels, and by implication a patient identifier and a treatment-arm column. Each patient belongs to one arm, and only that arm's outcome and label are filled in. Phenotype mix is similar across arms: adult onset 61.8%, juvenile onset 20.0%, late onset 18.2%. There are no missing values, but the data needs cleaning.

| Column | What it measures | Why it could matter |
|---|---|---|
| `ctg_repeats` | Number of CTG repeats in *DMPK* | The genetic cause; larger means more toxic RNA |
| `somatic_mosaicism_index` | How much the repeat varies between cells or tissues | Captures instability that the single repeat count misses |
| `family_history` (3 levels) | Inheritance pattern | Maternal transmission is linked to larger expansions |
| `age`, `age_at_onset`, `disease_duration` | Current age, age at first symptoms, years with disease | Timeline and accumulated damage. Arithmetically linked: age minus onset is duration |
| `phenotype` (3 levels) | Juvenile, adult or late onset | A category built from onset age; overlaps with it and with repeat count |
| `myotonia_severity` | Baseline muscle stiffness | The symptom mexiletine treats |
| `muscle_strength` | Baseline force | How much working muscle remains |
| `ck_level` | Creatine kinase, a blood marker of muscle damage | Plausible muscle marker; may or may not carry signal |
| `pr_interval` | ECG time for the electrical signal to cross the upper heart | Cardiac conduction; relevant to the disease, uncertain for these outcomes |
| `cognitive_function` | Thinking ability score | Half of the tideglusib outcome is cognitive |
| `insulin_resistance`, `bmi`, `albumin` | Sugar handling, body size, general nutritional state | Metabolic involvement; plausible but possibly noise |
| `cataracts` (yes/no) | Lens clouding | A disease feature unrelated to muscle outcomes |
| `sex`, `smoking_status` | Demographics and lifestyle | Generic covariates |

**The trap.** "Not every covariate carries signal. Some look plausible but are noise." Every column sounds medically relevant, so biological plausibility alone cannot decide. The data must select the covariate, and biology must then explain it.

## 6. The approach, step by step

The kickoff gives a six-step pipeline and says "each step protects the next one from bias". The steps below follow it, with detail on what to do and why.

### Step 1. Understand and clean

- Look at each covariate within each arm: range, skew, outliers, how much it varies.
- Look for correlated covariates and for the balance of responders and non-responders.
- **Clean.** Check for inconsistent category spellings, duplicate rows, numbers stored as text, impossible values (negative measurements, onset later than current age, duration that does not match age minus onset), and extreme values.
- **Rule:** decide every cleaning action from baseline columns and plausibility, never from the outcome, and write each decision into the code with its reason. Removing a patient because the outcome looks odd biases the result.
- Once the split is made, explore the training data only.

### Step 2. Split once, per arm, before modelling

- **Split each arm separately,** because outcomes are arm-specific.
- **Stratify by outcome,** so training and test sets have a similar mix. Use the responder label directly; for a continuous outcome, bin it first.
- **Fix the random seed** so the split can be reproduced.
- **Lock the test set** and use it only for the final evaluation.

Why: a model always looks good on the data it learned from. The test set stands in for future patients. If it influences any earlier choice, the reported accuracy is too optimistic. This is called leakage.

### Step 3. Preprocess only what the method needs

- Turn categories into numbers; scale numeric columns where the method requires it (penalised regression does, tree-based methods do not).
- Learn any transformation from the training set, then apply the same fitted transformation to the test set. Never refit on test data.

### Step 4. Select methods that fit the data

About 167 patients per arm and 18 covariates is a small problem. The slides say "simpler or penalised models suit small samples". Use a short list from each family (section 7).

### Step 5. Train

Fit and tune using the training set only. Tuning means choosing a model's settings; do it by cross-validation inside the training set, and keep the grid of settings small: "a large grid on a small arm mostly finds noise".

### Step 6. Evaluate once, then compare covariates

- Build **one scoreboard per arm** with every method scored on the same test patients.
- Show untuned and tuned ML side by side; add bootstrap intervals; do not over-read small gaps.
- **Compare the covariates, not only the scores.** Two methods can score alike and point to different covariates. Report what each selected.

### Beyond the six steps: where the clinical points are

- **Separate signal from noise.** Agreement across methods builds confidence; a covariate that only one method picks "needs a second look".

- **Interpret.** Connect each finding to a plausible clinical or biological mechanism.
- **Propose stratification or trial-design implications,** with clear, decision-relevant visuals. The slides list these three as "genuine innovation" beyond correct execution.

**Author's suggestions on choices the slides leave open.**

- **Outcome:** model the continuous outcome as primary and report the responder label alongside. The label is derived from the continuous value, so it carries less information; with about 40 test patients per arm, that matters.
- **Split ratio:** 75/25, giving roughly 125 training and 42 test patients per arm.
- **Noise checks:** add a column of pure random numbers as a negative control (anything ranked at or below it is noise); refit LASSO on many resamples of the training data and count how often each covariate is kept; and use Boruta, an established test of whether a covariate beats shuffled copies of the covariates in a random forest.
- **Calling a covariate "signal":** require most methods to agree and the resampling to be stable. Covariates picked only by the ML methods go to an "uncertain" list and are examined for a curved relationship or an interaction, which is where ML could add something.

## 7. The methods in plain words

The slides place methods on a line from interpretable to flexible, and note that the boundary between "covariate search" and "machine learning" is blurry by design: LASSO both selects covariates and predicts.

| | Covariate search | Machine learning |
|---|---|---|
| Core question | Which variables matter, and how? | Can we predict the outcome accurately? |
| Priority | Interpretability and inference | Predictive performance |
| Output | Coefficients, selected variables | Predictions, feature importance |
| Risk | Oversimplification | Black box |

### Covariate models

| Method | What it does | Watch out |
|---|---|---|
| Linear / logistic regression | Gives each covariate a weight; the weight is the effect size. Linear for a continuous outcome, logistic for a yes/no label | Assumes straight-line effects; sensitive to outliers |
| Stepwise selection | Adds or removes covariates one at a time using a fit criterion (AIC or BIC) | Unstable: a slightly different dataset can select a different set. "Don't treat the final model as the truth" |
| Best subset | Tries every combination of covariates and keeps the best of each size | Still overfits without a proper criterion; prefer BIC to $R^2$ |
| LASSO | Regression with a penalty that pushes weak coefficients to exactly zero, leaving a short list | May arbitrarily drop one of two correlated covariates |
| Ridge | Penalty that shrinks all coefficients but removes none | Does not select |
| Elastic net | A mix of the two: selects, and shrinks correlated covariates together | Needs cross-validation to set the penalty, like the others |
| Polynomial, splines, GAM | Let a covariate's effect bend instead of being a straight line | Too much flexibility overfits; harder to read |

| | | |
|---|---|---|
| PCA | Combines correlated covariates into a few summary components | Components are hard to interpret clinically and ignore the outcome |

### Machine-learning models

| Method | What it does | Watch out |
|---|---|---|
| Decision tree | A chain of yes/no questions on covariates; gives readable cut-offs | A deep tree memorises noise; limit the depth |
| Random forest | Many trees on random slices of the data, which then vote. Captures interactions without specifying them and ranks covariates | Harder to read than one tree |
| XGBoost | Small trees built one after another, each correcting the last. Often strong on tabular clinical data | Many settings; overfits quickly with few patients |
| SVM, K-nearest neighbours | A widest-margin boundary; or "patients like you" | Hard to explain; noise covariates mislead the neighbours |
| Neural network | Learned combinations of covariates | Least transparent; small arms limit what it can learn |

### Tools for seeing what an ML model used

- **Permutation importance.** Shuffle one covariate and see how much worse the predictions get. Correlated covariates can share the credit.
- **SHAP.** Splits each patient's prediction into a contribution from every covariate, with direction. "It explains the model, not biology. Importance does not prove causation."

## 8. Reading the results

### Metrics for a continuous outcome

| Metric | Meaning | Reading |
|---|---|---|
| RMSE | Typical size of the prediction error, penalising large misses more | Lower is better; in the outcome's own units |
| MAE | Average size of the prediction error | Lower is better; in the outcome's own units |
| $R^2$ | Share of the variation between patients that the model explains, relative to always predicting the test-set average | 1 is perfect, 0 is no better than the average, below 0 is worse than the average |

RMSE and MAE cannot be compared across arms because the outcomes use different units.

### Metrics for the responder label

| Metric | Question it answers |
|---|---|
| Recall | Of the true responders, how many did we find? |
| Precision | Of those we predicted to respond, how many did? |
| Accuracy | What share of all patients did we classify correctly? Misleading when responders are the minority: a model that predicts "nobody responds" can still score 70% |
| ROC-AUC | Does the model rank true responders above non-responders, across every cut-off? 0.5 is guessing, 1.0 is perfect. Unrelated to the pharmacokinetic AUC |

### How to answer the hackathon question from the outputs

- **If ML and covariate models score about the same and pick the same covariates:** the signal is simple and additive, and ML added nothing beyond confirmation. That is a legitimate answer.
- **If ML scores clearly better, or consistently picks a covariate the linear models miss:** look for a curved effect or an interaction between two covariates, confirm it with a simple plot, and say what ML revealed.
- **If ML scores worse:** small arms favour simple models; say so.

In every case, with about 40 test patients per arm "one patient moves the scores noticeably", so state the uncertainty and avoid crowning a winner on a small difference.

## 9. Pitfalls and likely judge questions

| Pitfall | How to avoid it |
|---|---|
| Leakage | Split first. Cleaning rules, scaling, selection and tuning use training data only. Score the test set once |
| Selecting noise | Require agreement across methods and stability under resampling; use a negative-control column |
| Correlated covariates | Report groups (for example "onset and genetic burden") when methods disagree on which member to pick |
| Over-reading scores | Report intervals; show untuned and tuned results; do not rank on small gaps |
| Wrong direction | Mexiletine: lower is better. The other two: higher is better. Say it on the slide |
| Accuracy alone | Always pair it with recall and precision, and state the cut-off |
| Importance read as cause | Describe covariates as predictors, and link them to mechanism as a plausible explanation |

**Two further cautions from the author.**

- **No placebo arm.** Each patient received one drug. The analysis can show who does well on a drug, but cannot separate "responds to this drug" from "would have done well anyway". A covariate found here is prognostic within the arm; showing it is predictive of drug benefit would need a control group.
- **Room to improve.** For outcomes measured as a change, patients who start worse often have more room to get better. When baseline severity predicts response, say that this can be partly a statistical effect and not only biology.

### Questions a judge may ask (one question, three minutes)

1. How did you separate real predictors from noise?
2. How did you split the data, and how do you know nothing leaked?
3. Why did you model the continuous outcome (or the label)?
4. Did ML add anything? Why or why not?
5. Why does this covariate make sense for this drug?
6. How would you use this to design the next trial?
7. How robust is the result to a different split?

## 10. Logistics and scoring

| When | What |
|---|---|

| | |
|---|---|
| Sun 11 to Tue 13 Oct | Teams work independently. Magnolia 1 is available on 12 and 13 October |
| Tue 13 Oct, 8:00 to 9:00 AM | Office hour in Magnolia 1 with the co-chairs and facilitator: non-technical questions and resource guidance. Questions may also be uploaded to the team folder at any time |
| **Tue 13 Oct, 11:59 PM ET** | **Slide deck and all code files uploaded to the Team 1 SharePoint folder.** Nominate one representative to email the chairs that the slides are final. The chairs merge all decks into one presentation |
| Wed 14 Oct, 9:00 to 10:30 AM | Ten teams present: 5-minute talk plus 3-minute Q&A with one judge question. Judges score live. At least one member presents |
| Wed 14 Oct, 4:30 to 5:00 PM | Winners and any honourable mentions announced at the closing session |

| Category | Points | What judges look for |
|---|---|---|
| Clinical | 10 | Key predictors per arm; real signal separated from noise; clinical interpretation and application |
| Technical | 10 | Modelling workflow; unbiased split and justified method; assessment metrics |
| Communication | 10 | Clear story within the time limit; supportive visuals; confident Q&A; meaningful conclusion |

**Submission checklist from the slides:** slides tell one story (workflow, key covariates, interpretation, conclusion); code files are complete and reproduce the results; split, noise handling and metrics are explained and justified; key covariates are separated from noise; AI tools used are reported with the purpose of each.

**Rules.** Any platform or language. LLM chatbots and AI agents are welcome, but the team must understand every step of the final analysis, report which AI tools were used and for what, and remains accountable for all results. The co-chairs and facilitator support teams and do not take part in judging. Prize: medals and a $100 gift card for the winning team; certificates for all participants.

**Team 1** (roster marked "subject to change"): Euibeom Shin, Luong Vuong, Vijay Ivaturi, Ming Gao, Wenzhe Ma, Bei Zhao.

## 11. Glossary

| Term | Meaning |
|---|---|
| Covariate | A baseline patient characteristic used to explain or predict the outcome |
| Stratification | Sorting patients into groups expected to behave alike |
| CTG repeat, *DMPK* | The repeated DNA triplet, and the gene that carries it |
| Somatic mosaicism | Different repeat lengths in different cells and tissues of one person |
| Anticipation | Earlier and more severe disease in successive generations |
| Splicing; MBNL | The editing of RNA messages; the editor protein trapped in DM1 |
| Myotonia | Delayed muscle relaxation |
| QMT; vHOT | Quantitative muscle testing, a strength score; video hand opening time, a myotonia measure |
| PR interval; CK | An ECG measure of conduction time; creatine kinase, a blood marker of muscle damage |
| GSK3-beta | The enzyme tideglusib blocks |
| Training / test set | Patients the model learns from / patients held back to check it |
| Leakage | Test-set information influencing model building, which inflates scores |

| | |
|---|---|
| Cross-validation | Repeatedly holding out part of the training data to choose settings |
| Overfitting | Learning noise in the training data that does not hold in new patients |
| Prognostic / predictive | Predicts outcome regardless of treatment / predicts benefit from a specific treatment |

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Sources: ACoP 2026 Hackathon kickoff slide deck and session notes, 11 October 2026. The dataset is simulated and does not describe real patients or approved therapies. Green boxes are the author's reading or suggestions and are not from the organizers. Prepared with Claude (Anthropic).